

AML HOOK

Modular Compliance Layer for Uniswap v4

Use Case — Exploit Detection, Propagation and N-Hop Decay

1. Overview

AML Hook is a compliance layer deployed natively as a Uniswap v4 hook. It intercepts every swap at beforeSwap and afterSwap, applies a ternary risk decision, and emits a structured on-chain event that constitutes the operator's audit trail. The hook operates without interrupting the normal swap execution path for clean addresses.

This document describes a complete three-wallet scenario that exercises all three decision outputs of the hook in a single run: full block (revert), punitive dynamic fee, and proportional dynamic fee. The scenario is grounded in an exploit cash-out attack followed by two-hop fund propagation through intermediary wallets.

Actors

Wallet	Role	Initial Score
Wallet A	Exploit attacker. Drains an external lending protocol and attempts to use the pool to convert stolen USDC into ETH.	100 (exploit confirmed)
Wallet B	First-hop intermediary. Receives stolen funds directly from Wallet A via a peer-to-peer (P2P) transfer outside the pool.	65 (1-hop decay)
Wallet C	Second-hop intermediary. Starts with a clean history, later receives a portion of funds from Wallet B.	0 → 42 (2-hop decay)

The pool is configured as a Real World Asset (RWA) pool on Uniswap v4, with AML Hook attached and dynamic fee permissions enabled. The off-chain scoring keeper monitors transfer events continuously and writes updated scores on-chain before any swap is attempted.

2. Risk Scoring Model

2.1 Ternary Decision Logic

Score Range	Hook Response	Fee Applied	Regulatory Basis
0 – 30	Allow	Standard (0.30%)	No risk indicators. Normal execution.
31 – 70	Allow with punitive fee override (lpFeeOverride)	Dynamic (3% – 8%)	Suspected contamination, not yet confirmed. Enhanced Due Diligence (EDD) equivalent. Creates economic friction without blocking.
71 – 100	Revert	N/A	Confirmed exposure: exploit cluster, OFAC match, or direct link to sanctioned entity. No discretion.

2.2 N-Hop Decay Formula

Contamination from a tainted source propagates to downstream wallets with a mathematically decreasing weight.

The formula applied by the off-chain keeper is:

$$\text{derived_score} = \text{origin_score} \times (\text{decay_factor} ^ \text{hops}) \times \text{exposed_proportion}$$

Parameter	Value Used	Description
decay_factor	0.65	Contamination weight retained per hop. Industry-aligned: meaningful signal up to 3 hops, negligible beyond 4.
exposed_proportion	1.0 (full transfer)	Fraction of the receiving wallet's balance that originates from the tainted source.
origin_score	100 (Wallet A)	Score of the originating tainted wallet, as written by the keeper after exploit detection.

Calculated scores for this scenario:

Wallet	Hops from Source	Calculation	Resulting Score	Hook Tier
A	0 (source)	Detected directly via on-chain exploit event	100	Revert
B	1	$100 \times 0.65^1 \times 1.0 = 65$	65	Punitive fee (8%)
C (after receiving from B)	2	$100 \times 0.65^2 \times 1.0 = 42$	42	Proportional fee (3%)

3. Full Execution Sequence

Step 0 – Baseline: Clean Swap (Wallet C, pre-contamination)

Before any exploit occurs, Wallet C executes a standard RWA swap in the pool. At this point C has no risk history.

Event	Detail
Actor	Wallet C
Action	Swap USDC → RWA token at standard pool conditions.
beforeSwap – Layer 1	OFAC screening: no match.
beforeSwap – Score read	Keeper score: 0. Below 30 threshold. No fee override applied.
Execution	Swap executes at standard fee (0.30%).
afterSwap – Event emitted	{ address: C, score: 0, decision: ALLOW, fee: 0.30%, amount: X, timestamp: T0 }
Hook output	ALLOW – standard fee. No friction for legitimate operators.

Step 1 – Exploit Detection: Direct Attack (Wallet A)

Wallet A executes an exploit against an external lending protocol, extracting \$10M USDC. The attacker immediately attempts to swap those funds in the AML Hook pool to convert them into ETH before Circle freezes the USDC address.

The off-chain keeper monitors transfer events and mempool activity. It detects the exploit event on-chain, traces the USDC outflow to Wallet A, and writes score 100 before the swap transaction is confirmed.

Event	Detail
Actor	Wallet A
Action	Swap \$10M USDC → ETH.
Keeper — prior to swap	Exploit event detected on-chain. Score 100 written to oracle for Wallet A.
beforeSwap — Layer 1	OFAC screening: no match (designation lag, list not yet updated).
beforeSwap — Score read	Keeper score: 100. Threshold 71 exceeded. revert() executed atomically.
Execution	Transaction reverts. No funds move. No liquidity provider (LP) exposure.
afterSwap	Not reached. Revert occurs in beforeSwap.
Hook output	REVERT — exploit cluster confirmed. Block precedes OFAC designation by hours.

Step 2 — First-Hop Propagation: Wallet A → Wallet B (P2P Transfer)

Blocked at the pool, Wallet A routes the stolen funds via a peer-to-peer transfer directly to Wallet B, an intermediary address with no prior risk history. The transfer occurs outside the Automated Market Maker (AMM), at the ERC-20 token level.

The keeper detects the inbound transfer to Wallet B, traces its origin to Wallet A (score 100), and applies the decay formula: $100 \times 0.65^1 \times 1.0 = 65$. Score 65 is written to the oracle for Wallet B before any swap is attempted.

Step 3 — First-Hop Swap Attempt (Wallet B, Score 65)

Event	Detail
Actor	Wallet B
Action	Attempts to swap USDC → ETH in the AML Hook pool.
beforeSwap — Layer 1	OFAC screening: no match.
beforeSwap — Score read	Keeper score: 65. Falls in tier 31–70. No revert. lpFeeOverride applied: 8%.

Execution	Swap executes. Fee of 8% deducted. Net proceeds to Wallet B reduced significantly.
afterSwap — Event emitted	{ address: B, score: 65, decision: FEE_OVERRIDE, fee: 8.00%, hop_distance: 1, origin: A, timestamp: T2 }
Hook output	PUNITIVE FEE — direct contamination from exploit source. Economic penalty applied. Full audit trail recorded.

Step 4 — Second-Hop Propagation: Wallet B → Wallet C (P2P Transfer)

Wallet B transfers a portion of the remaining funds to Wallet C. The keeper detects the transfer, traces the contamination chain ($A \rightarrow B \rightarrow C$), and calculates the two-hop score: $100 \times 0.65^2 \times 1.0 = 42$. Score 42 is written to the oracle for Wallet C, overwriting the prior clean score of 0.

Step 5 — Second-Hop Swap Attempt (Wallet C, Score 42)

Event	Detail
Actor	Wallet C
Action	Attempts to swap USDC → ETH in the AML Hook pool.
beforeSwap — Layer 1	OFAC screening: no match.
beforeSwap — Score read	Keeper score: 42. Falls in tier 31-70. lpFeeOverride applied: 3% (proportional to lower contamination).
Execution	Swap executes. Reduced fee reflects two-hop distance from exploit source.
afterSwap — Event emitted	{ address: C, score: 42, decision: FEE_OVERRIDE, fee: 3.00%, hop_distance: 2, origin: A, timestamp: T4 }
Hook output	PROPORTIONAL FEE — two-hop contamination. Penalty decays with distance. Audit trail maintained.

4. Sequence Summary

Step	Actor	Action	Score	Hook Decision	Fee
0	C	Standard swap (pre-contamination)	0	ALLOW	0.30%

1	A	Direct swap attempt with exploit funds	100	REVERT	N/A
2	A → B	P2P transfer outside pool. Keeper writes score 65 to Wallet B.	—	Off-chain keeper update	—
3	B	Swap attempt with 1-hop contaminated funds	65	FEE OVERRIDE (punitive)	8.00%
4	B → C	P2P transfer outside pool. Keeper writes score 42 to Wallet C.	—	Off-chain keeper update	—
5	C	Swap attempt with 2-hop contaminated funds	42	FEE OVERRIDE (proportional)	3.00%

5. On-Chain Audit Trail

Every hook intervention emits a structured event via afterSwap. These events are indexed by The Graph and constitute the immutable audit record that the pool operator presents to regulators (Financial Crimes Enforcement Network, FinCEN; Markets in Crypto-Assets Regulation, MiCA; Office of Foreign Assets Control, OFAC) to demonstrate transaction-level due diligence. They also serve as the factual input for any Suspicious Transaction Report (STR) filed by the operator's compliance function.

Each event record contains:

- The wallet address screened.
- The risk score at the time of the swap.
- The decision taken (ALLOW, FEE_OVERRIDE, or REVERT).
- The fee applied, or the revert basis.
- The hop distance from the contamination source, where applicable.
- The origin wallet traced as the contamination source.
- The transaction amount and timestamp.

No existing compliance hook on Uniswap v4 produces this record. Binary allowlist models either permit or block, and emit no structured data on the decision rationale.

6. Regulatory Basis

Hook Behavior	Regulatory Principle
Score 71–100: unconditional revert	OFAC mandatory blocking obligation. Financial Action Task Force (FATF) Recommendation 6: targeted financial sanctions, no discretion.
Score 31–70: punitive fee override	FATF Recommendation 10: Enhanced Due Diligence for higher-risk situations. Risk-based approach: not all risk justifies a block; friction and monitoring are the appropriate response at intermediate risk.
N-hop decay scoring	FATF Virtual Assets Red Flag Indicators (2020): indirect exposure to illicit funds constitutes a risk indicator. The system must trace fund origin, not just direct counterparty.
Immutable on-chain event log	FATF Recommendation 11: record retention, minimum five years. The audit trail is on-chain and non-modifiable.

8. Conclusion

This scenario demonstrates the complete decision space of AML Hook in a single execution run. A direct attack by the exploit source triggers an immediate revert. An attempt to route contaminated funds through intermediary wallets is detected and penalized with mathematically decaying fees proportional to hop distance. Every decision is recorded on-chain with full traceability.

The two attack vectors covered — direct exploit cash-out and multi-hop fund propagation — represent the primary evasion strategies used in DeFi exploits. AML Hook addresses both at the execution layer, without requiring the attacker to appear on any external list, and without introducing latency into the normal swap path for clean addresses.